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Alimentación, carga y batería

Archivo: power.kicad_sch

ESP32-S3

Archivo: mcu.kicad_sch

GNSS UM980 y antena

Archivo: gnss.kicad_sch

microSD, IMU y panel

Archivo: io.kicad_sch

Placa principal exploratoria de TresVizo MeridianV. Nada de esto se ha fabricado ni probado.

Variante 1S por defecto. Variante 2S: R105 = 8.2k (PROG), U103 = MAX17049G+T10, R115 sin montar,

R117 montado (VDD del medidor desde +3V3) y R119 (EN del buck, abajo) = 3.9k.

JLCPCB 4 capas JLC04161H-7628, 50 x 76 mm

UM980 + ESP32-S3-MINI-1-N4R2 + BMI088 + BQ25798 (1S; 2S por variante)

Exploratorio: no fabricado ni probado

TresVizo

Sheet: /

File: tresvizo-main.kicad_sch

Title: TresVizo MeridianV – placa principal

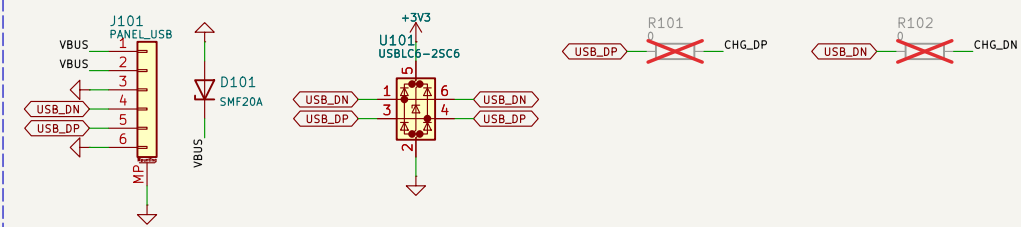
Size: A4

Date: 2026-10-04

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[illegible]

Cargador BQ25798RQMR (buck-boost NVDC, 1S por defecto; 2S por PROG)

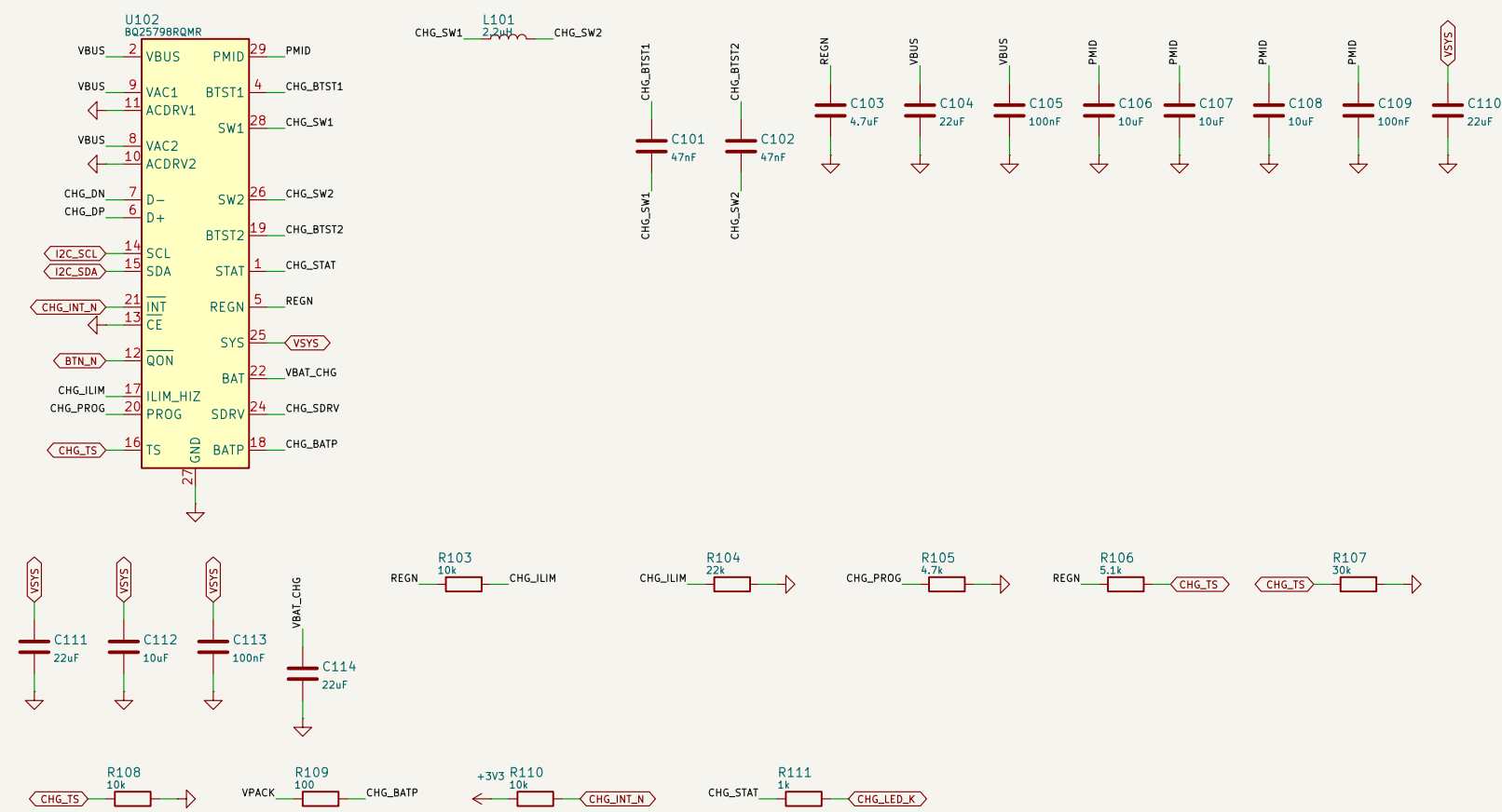
Pin	Signal	Pin	Signal
1	CHG_STAT	13	INT
2	VBUS	14	I2C_SCL
3	BTST1	15	I2C_SDA
4	CHG_BTST1	16	CHG_TS
5	REGN	17	BTIN_N
6	CHG_DP	18	CHG_BATP
7	CHG_DN	19	CHG_BTST2
8	VAC2	20	CHG_PROG
9	VAC1	21	CHG_INT_N
10	ACDRV1	22	VBAT_CHG
11	ACDRV2	23	CHG_ILIM
12	QON	24	CHG_SDRV
13	CE	25	SYS
14	SCL	26	CHG_SW2
15	SDA	27	GND
16	CHG_TS	28	CHG_SW1
17	BTIN_N	29	PMID
18	CHG_BATP		
19	CHG_BTST2		
20	CHG_PROG		
21	CHG_INT_N		
22	VBAT_CHG		
23	CHG_ILIM		
24	CHG_SDRV		
25	SYS		
26	CHG_SW2		
27	GND		
28	CHG_SW1		
29	PMID		

External Components:

- Capacitors:** C111 (22uF), C112 (10uF), C113 (100nF), C114 (22uF), C101 (47nF), C102 (47nF), C103 (4.7uF), C104 (22uF), C105 (100nF), C106 (10uF), C107 (10uF), C108 (10uF), C109 (100nF), C110 (22uF).
- Resistors:** R103 (10k), R104 (22k), R105 (4.7k), R106 (5.1k), R107 (30k), R108 (1k), R109 (10k), R110 (10k), R111 (1k).

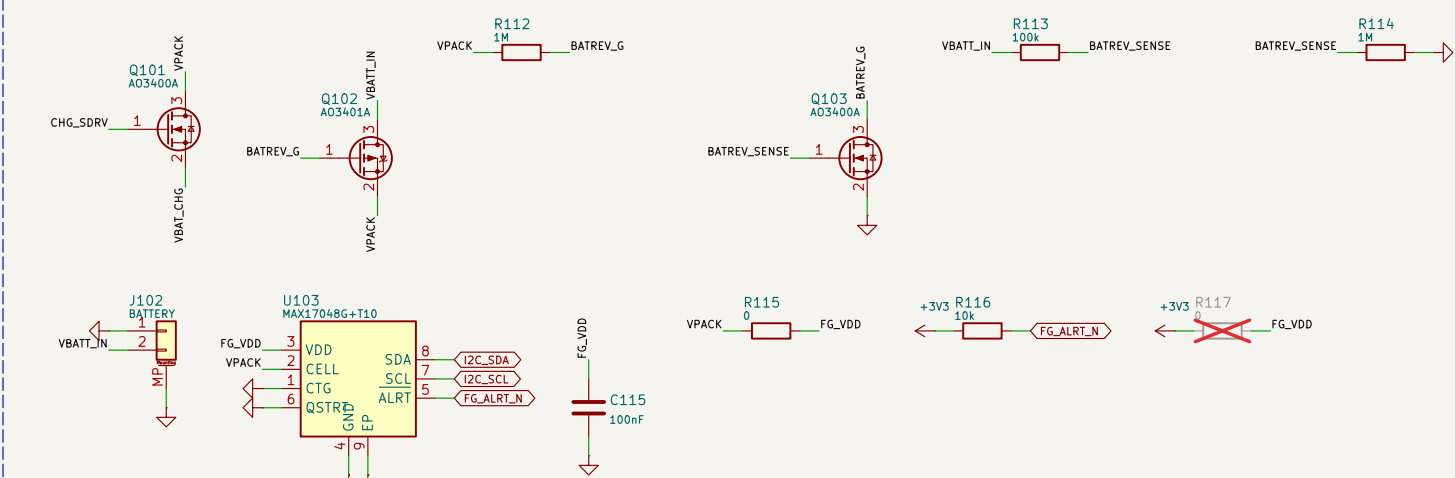
Connections:

- VSYS to C111, C112, C113, and VSYS.
- VBAT_CHG to C114.
- REGN to R103, R104, R105, R106, and R107.
- CHG_ILIM to R103 and R104.
- CHG_PROG to R105.
- CHG_TS to R106 and R107.
- CHG_TS to R108.
- VPACK to R109.
- +3V3 to R110.
- CHG_STAT to R111.

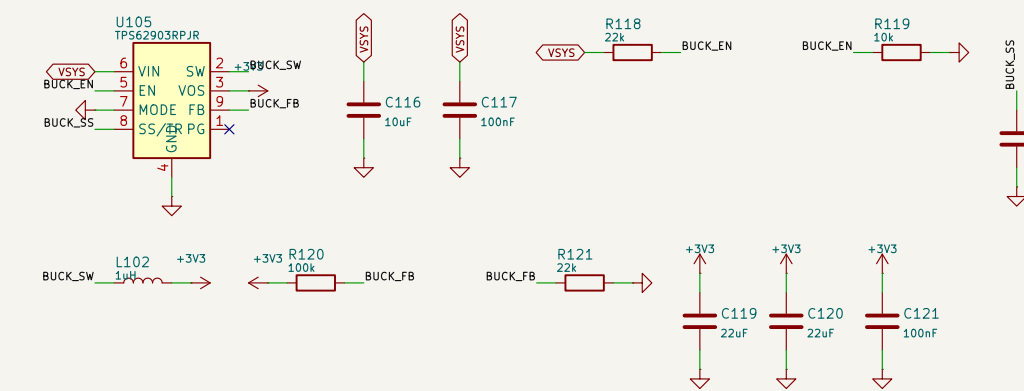


Bateria: polaridad inversa, FET de apagado (ship) y medidor

The diagram illustrates a battery management system (BMS) for a 12V battery. The system includes a 12V battery (U102), a BMS IC (U103, MAX31248G+T10), a MOSFET (M101), and various status signals (I2C_SDA, I2C_SCL, FG_ALERT_N). The BMS IC is connected to the battery terminals (VBATT_N, VBATT_P) and the battery pack (VPACK). The BMS IC also controls a MOSFET (M101) and provides status signals (I2C_SDA, I2C_SCL, FG_ALERT_N). The BMS IC is connected to the battery pack (VPACK) and the battery (U102). The BMS IC is connected to the battery pack (VPACK) and the battery (U102). The BMS IC is connected to the battery pack (VPACK) and the battery (U102).



3.3 V: TPS62903 (3–17 V, 3 A, modo 100 %) [\[1\]](#)



Riel conmutado del UM980

U106
TPS22910DCKR

+3V3

GNSS_SW

GNSS_PWR_EN

GNSS_SW

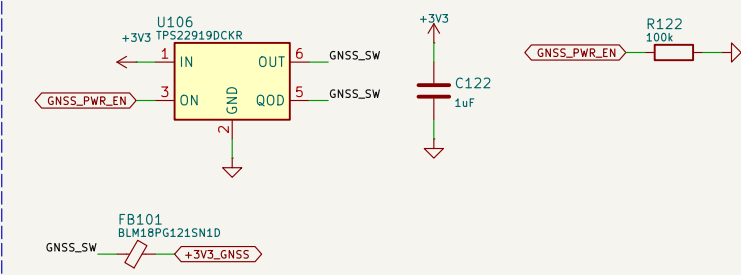
GND

C122
1uF

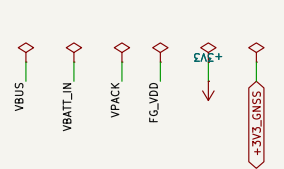
R122
100k

FB101
BLM18PG1215N1D

+3V3_GNSS

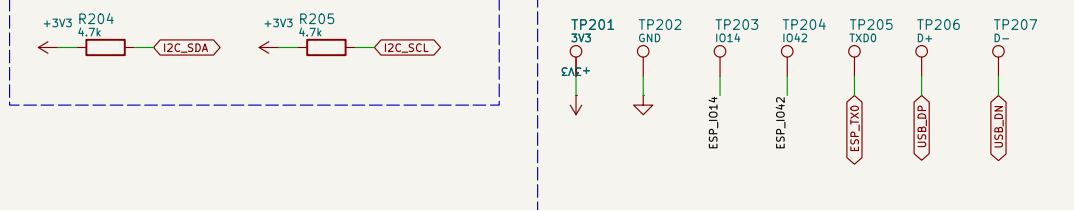
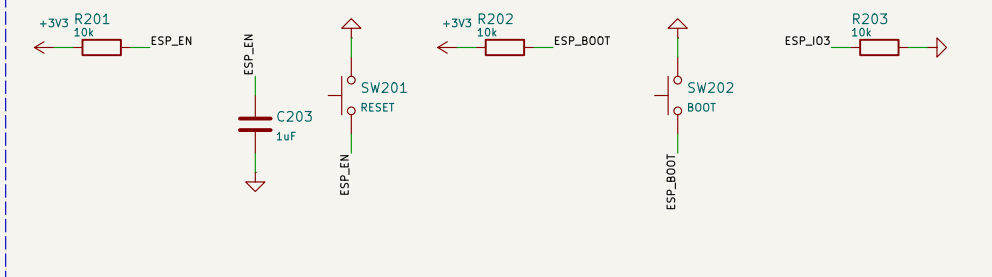
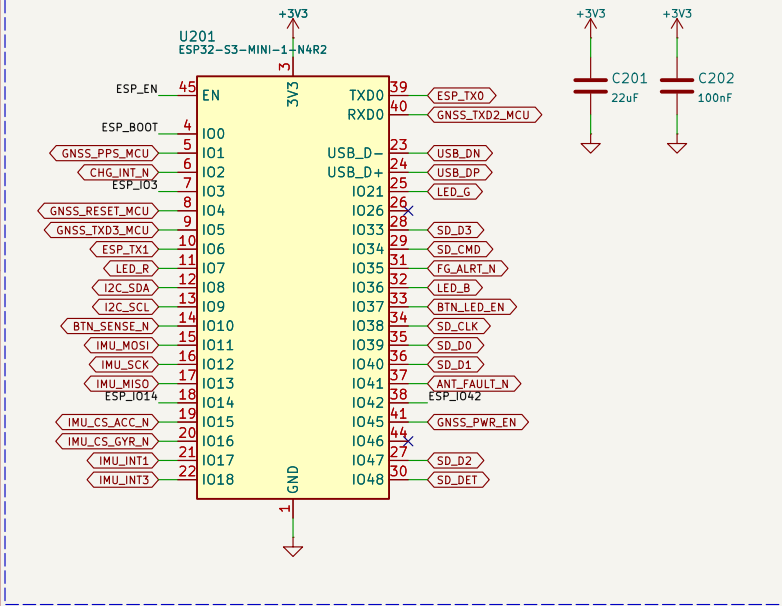


Banderas de alimentación (ERC)

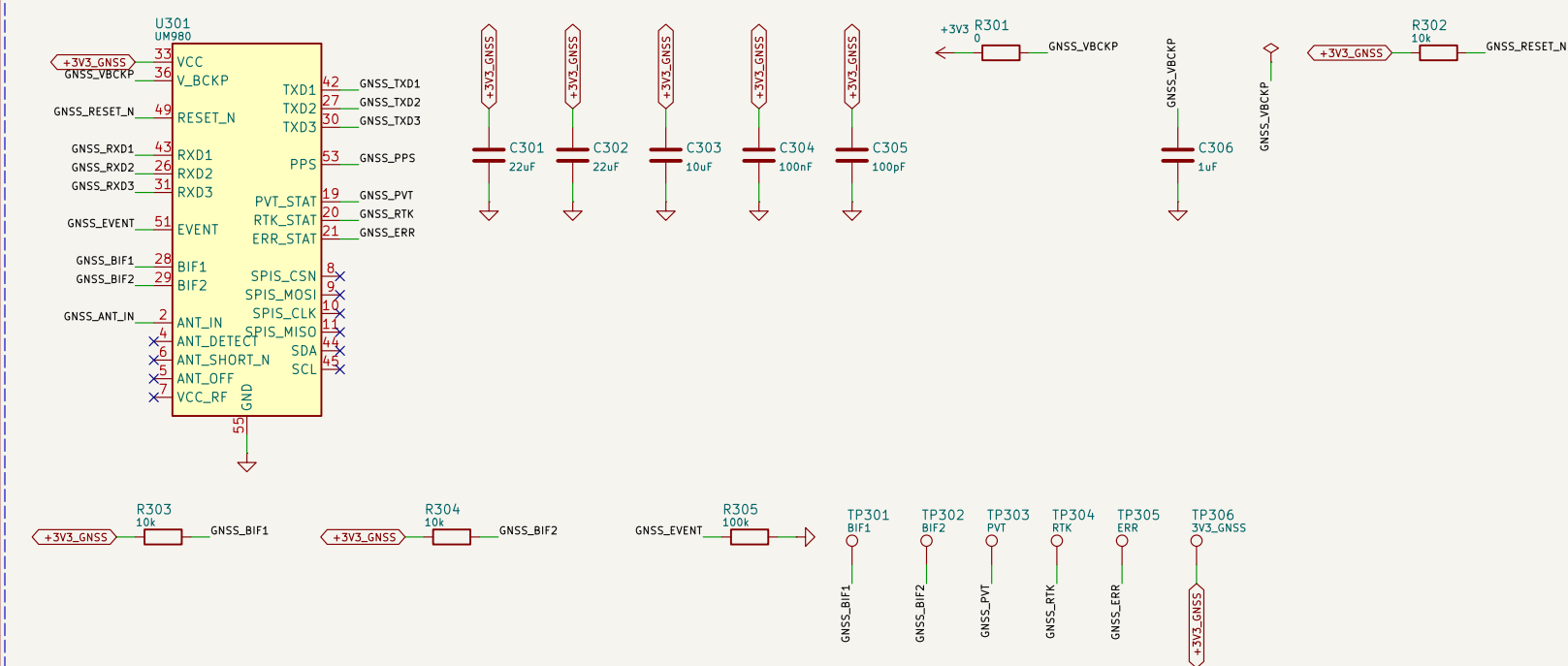


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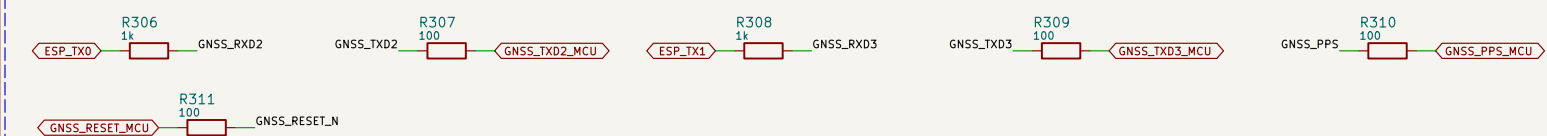
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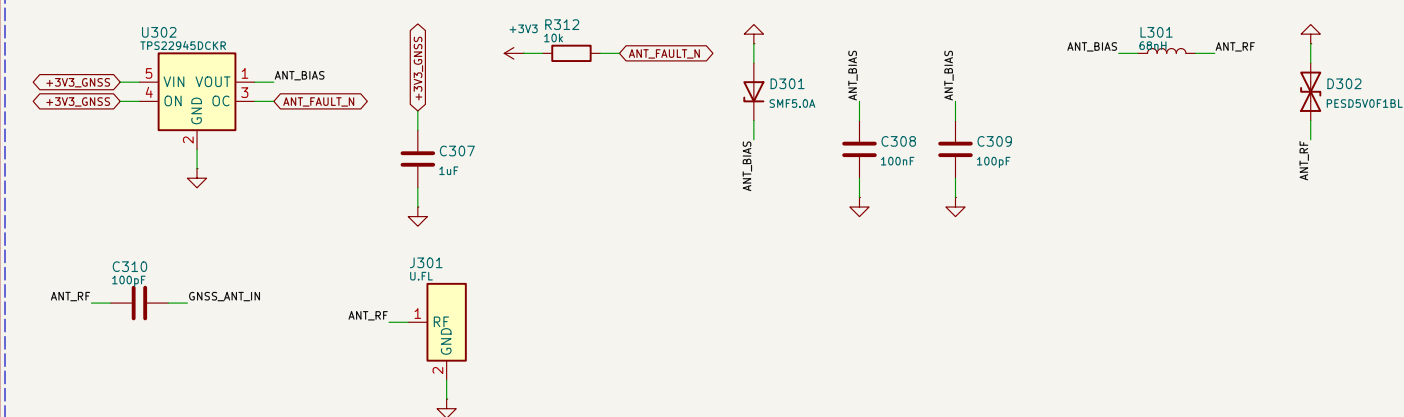
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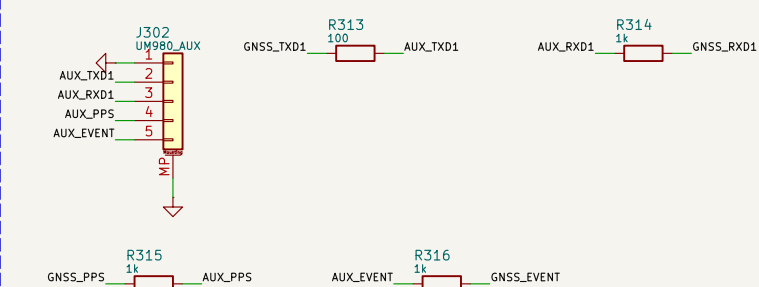
Resistencias serie (100 ohm desde el UM980, 1 k hacia el UM980)



Antena activa: límite de corriente, bias-T y u.FL (manual Unicore, fig. 3-2)



Puerto auxiliar UM980 (UPrecise, PPS, EVENT)



JLPCPB 4 capas JLC04161H-7628, 50 x 76 mm
UM980 + ESP32-S3-MINI-1-N4R2 + BMI088 + BQ25798 (1S; 2S por variante)
Exploratorio: no fabricado ni probado

TresVizo

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